Nodal-Precession Phasing for Multi-Object Debris Removal in Low Earth Orbit

An Audited and Reproducible Re-evaluation of Waiting below the Targets, with Exact Optimiser Checks, a Servicer Design and Cost Assessment, and a Comparison with Removal Demonstrations

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Abstract

Massive derelict rocket bodies in a few low Earth orbit bands dominate the future fragment supply, and a single servicing spacecraft could visit many of them if it lets the J_2 nodal precession rotate the orbit planes into line instead of paying for plane changes. This paper re-derives, tests and re-runs an earlier draft study of that idea on one frozen public snapshot (Celestrak catalogue and element sets of 18 August 2026, common epoch 15:02:52 UTC, hashes published) and on the recovered 2025 inputs, and it adds the subsystem design, cost and policy view that a real servicer needs. The draft's catalogue counts are reproduced exactly from the 2025 files: the 1,288 "actionable" objects are the non-operational payloads and rocket bodies at 600–1,500 km mean altitude in a March 2025 catalogue, 193 of which had already decayed, and the 593 objects in 40 sub-clusters follow once April 2025 node angles are joined; on the 2026 snapshot the same rule restricted to objects in orbit gives 1,151. The "lower is better" note rests on a Hohmann cost that is not symmetric, and the planner's scalarisation weight was never tuned. We prove that the drift-rate gain of lowering exceeds that of raising for every offset ($R(x) > 1$, series $1 + \frac{9}{2}x + \frac{81}{8}x^2 + \dots$), that the cost ratio is $1 + \frac{3}{2}x$ and that the net gain per ΔV is $1 + 3x$, and we give a decision rule, an eccentricity reversal and error bounds. Against an exact dynamic-programming optimum on 32 small instances, a tuned beam search is 1.9% (95% CI 0.6–3.4) from optimal, while the draft's configuration is 10.3% from it. On the 159-object SL-8 pool the median tour reaches 29 objects at 2 km/s and 7 yr, and 67 at 5 km/s and 15 yr, compared with 9 and 19 for impulsive plane changes and 6 and 12 for waiting without an altitude change. Cowell and SGP4 checks close the node to $\leq 0.05°$. Of ten novel ideas screened, only a parked carrier with expendable kits is supported (4.5–6 times faster kit placement than a serial tour). Every flown or planned removal demonstration handles one object; a chemical carrier parked 400 km below the SL-8 cluster releases 16 kits within about 310 days of launch, but kits that carry their own de-orbit propellant raise its launch mass from 3.3 t to 4.8–5.9 t. A parametric estimate after the SMAD cost models puts this 16-kit mission at 566 M USD (2024 dollars; 10–90%: 396–736 M), or about 35 M USD per object and 25,000 USD per kilogram removed, with most of the cost in development rather than in operations or launch. Electric propulsion saves propellant only on the long visit-only tour, where it needs 1.1 to 3.1 demonstrated thruster lifetimes. The main drawback of the approach is that its advantage is bought with years of mission time, and we claim no new astrodynamics.

Nomenclature

Orbit and Perturbation Parameters	
a, e, i	semi-major axis [km], eccentricity, inclination [deg]
p	semi-latus rectum, $a(1-e^2)$ [km]
n	mean motion [rad/s]
μ, R_E, J_2	Earth gravitational parameter, equatorial radius, second zonal harmonic
Ω, $\dot{\Omega}$	right ascension of the ascending node (RAAN) [deg], secular nodal rate [deg/day]
K	nodal-rate constant, $\frac{3}{2}J_2R_E^2\sqrt{\mu}$
Phasing Parameters	
x	relative altitude offset, $\Delta a/a_T$
$R(x)$	ratio of nodal-rate gain for lowering to that for raising
g, g^*	nodal gap [deg], gap threshold of the decision rule [deg]
h	drift-orbit depth below (or above) the target [km]
Δt	dwell time to close a nodal gap [d]
Mission Parameters	
ΔV	velocity change [m/s]
v, v_c	orbital speed, circular speed [m/s]
I_{sp}	specific impulse [s]
C_dA/m	ballistic parameter [m^2/kg]
$w_{\Delta V}$, w_t	planner scalarisation weights on ΔV and time
K (kits)	number of expendable kits placed by the carrier
Servicer Parameters	
P_e, P_d, P_{sa}	load in eclipse and in daylight, solar-array output required [W]
T_e, T_d	eclipse and daylight duration per orbit [min]
X_e, X_d	power-path efficiencies in eclipse and in daylight
I_d, L_d	inherent degradation and life degradation of the solar array
C_r, DOD	battery capacity [W h], depth of discharge
λ	series failure rate of the servicer [1/yr]
h_p	perigee altitude after a de-orbit burn [km]
L, S	learning-curve factor, learning-curve slope [%]
σ	standard deviation of a cost estimate [%]

1 Introduction

Collision-risk studies since Kessler and Cour-Palais [31] and the long-term projections of the space agencies [37–40] agree that massive intact derelicts, above all rocket bodies concentrated in a few inclination–altitude bands, dominate the future fragment supply [1, 43]. Active removal is expensive per object, so one launch should remove many. The obstacle is the orbit plane. Objects in one band share inclination but not right ascension of the ascending node (RAAN), and a direct plane change of $\Delta\Omega = 1^\circ$ at $i = 83^\circ$ costs 127.5 m/s at the SL-8 orbital speed; Braun et al. [4] identify plane change as the main obstacle to multi-target removal. The remedy, already used in constellation deployment [11, 32, 42] and in debris tours [6, 24, 49], is to let the secular J_2 drift of the node do the work: the spacecraft waits on an orbit whose nodal rate differs from that of the target.

This paper started from a draft study that proposed waiting *below* the targets (lowering) and reported tours of 24–53 objects. However, the draft's numbers were never independently

reproduced, its analytic claims were not checked, and its planner was never compared with an exact optimum. The goal of this paper is to address this gap. This study turns the draft into a reproducible evaluation to a publishable standard, and tests by calculation whether further ideas help. The contributions are evidential, not conceptual (Section 12.2):

1. an independent engine that reproduces the draft's tour exactly, and an audit of every analytic claim and pipeline stage (Section 3);
2. a frozen, hash-pinned snapshot that resolves the catalogue-count discrepancies;
3. exact and sharp statements on lowering versus raising (Section 4);
4. an exact-optimum validation of the sequencing heuristics and a correction of the planner's weight (Section 6);
5. fair baselines, high-fidelity verification, Monte Carlo uncertainty, a caveated collision-risk benefit and a screen of ten novel ideas;
6. an exact reproduction of the draft's 2025 catalogue pipeline from the original inputs, which explains every count it reported (Section 3.1);
7. a servicer design at the level SMAD expects for a mission concept [35], placed against the flown and planned removal demonstrations, with propulsion, power, mass, communications, reliability, de-orbit kit, timeline and cost estimates, and the regulatory context (Sections 10 and 11).

In-orbit servicing and removal have moved from studies to flight in the last decade. RemoveDEBRIS tested a net and a harpoon on its own CubeSat targets in 2018 [3, 13], ELSA-d released and recaptured its own magnetically prepared client in 2021 [3], ADRAS-J approached a non-cooperative upper stage in 2024 [27, 28], and GEO life-extension servicers have docked to operational clients since 2020 [3]. ClearSpace-1 [15, 16] and ADRAS-J2 [27] are to capture one object each later this decade. All of these handle one object per mission. The question of what a multi-object tour would look like as a spacecraft, and what it would cost, is therefore the natural next step for this paper.

The rest of the paper is structured as follows. Section 2 develops the approach: the secular nodal-rate and cost models, the frozen snapshot, the leg and tour model, the solvers, the baselines and the uncertainty and risk models. Section 3 reports the audit of the draft, Section 4 the exact results on lowering versus raising, and Sections 5 and 6 the validation of the numerical model and of the optimiser. The results are then presented for the fair baselines, the headline missions and the collision-risk benefit (Section 7), followed by uncertainty and sensitivity (Section 8) and the screen of novel methods (Section 9). Section 10 turns the tour into a servicer design and compares it with the demonstrations, and Section 11 gives the regulatory context. The discussion covers the implications, the novelty assessment, the sources of error and model limitations, and the objections a reviewer would raise (Section 12). The paper concludes with the main findings and the work that remains.

2 Approach

The goal of this study is to quantify how many objects a single chaser can visit in one band when the plane changes are done by J_2 nodal drift, and how that compares with the alternatives at equal budget. The analysis links

1. a secular nodal-rate model and exact transfer costs (Section 2.1),

2. a frozen catalogue snapshot and a leg model that turns each target visit into a transfer, a dwell and a final transfer, and
3. a tour optimiser checked against an exact dynamic program,

to compute the number of objects reached, the ΔV, the mission time and the propellant mass, together with their uncertainty.

2.1 Secular nodal rate and transfer costs

Secular nodal rate. For semi-major axis a, eccentricity e, inclination i and $p = a(1-e^2)$, the J_2 secular rate is [5, 33, 34]

$$\dot{\Omega} = -\tfrac{3}{2} J_2 n \Big(\frac{R_E}{p}\Big)^2 \cos i, \qquad n = \sqrt{\mu/a^3}, \tag{1}$$

where n is the mean motion, μ the Earth's gravitational parameter and R_E its equatorial radius. For $e = 0$ the rate reduces to $\dot{\Omega} = -K \cos i \; a^{-7/2}$, with $K = \frac{3}{2} J_2 R_E^2 \sqrt{\mu}$. A target whose rate is higher than the chaser's, in the direction of the gap, is reached after $\Delta t = g/|\dot{\Omega}_c - \dot{\Omega}_t|$ for a gap g. For absolute rates we use the second-order closed form (terms in J_2^2 and J_4) with the SGP4 constants, which reproduces SGP4's own mean nodal rate to 0.08%; Eq. (1) alone is 0.29% high on the SL-8 pool. However, only rate *differences* matter for phasing, so this common-mode bias cancels.

Costs. Hohmann transfers use the exact two-burn expression. A node error of θ_Ω at inclination i costs $2v \sin(\frac{1}{2}\vartheta)$, with $\sin\frac{1}{2}\vartheta = \sin i \, \sin\frac{1}{2}\theta_\Omega$ (equal inclinations). Drag make-up is computed with a tabulated exponential atmosphere and a ballistic parameter $C_d A/m = 0.01\,\mathrm{m}^2/\mathrm{kg}$ (stated, unverified). Propellant follows the rocket equation for two stated designs, both of which are assumptions: a storable bipropellant design ($I_{sp} = 320\,\mathrm{s}$, 700 kg dry) and a Hall-thruster design (1800 s, 900 kg dry).

2.2 Frozen snapshot and pools

The analysis uses *snapshot v3-2026-08-18*, which consists of the Celestrak SATCAT csv (70,292 rows) and the GP element sets for five pools, fetched by the original study on 18 August 2026 from public, credential-free sources; the SHA-256 hashes are in `data/SNAPSHOT.json`. The snapshot was not re-downloaded, because the draft's tours are defined on it and a new epoch would break the comparison. All elements are propagated with SGP4 to one common epoch (2026-08-18T15:02:52.746Z for the SL-8 set; TLE age median 0.14 d, maximum 0.87 d). The pools are SL-8 rocket bodies ($i \approx 82.9°$, 950–1,010 km, 159 objects), sun-synchronous objects (135), Iridium-class objects (28) and SL-16 rocket bodies (20). Nodal rates are evaluated on *mean* elements. The draft used osculating elements, which differ in a by 6.4 km (1σ), a difference comparable to the pool's own spread of 7.5 km. The original 2025 inputs of the draft (two public catalogue files of March and April 2025, element sets of April 2025 and the outputs of its MATLAB and Python stages) were recovered from the author's records and copied read-only into `data/orig2025/`; they are used only to reproduce the draft's counts (Section 3.1).

2.3 Leg model and tour problem

A leg takes the chaser from object c (in its plane) by a Hohmann transfer to a circular drift orbit $a_d = a_t \mp h$ below (or above) target t. The chaser then dwells until the secular nodal gap closes, and finally makes a combined Hohmann and inclination transfer to

t, plus a fixed proximity-operations allowance (15 m/s, stated). The depths considered are $h \in \{25, 35, 50, 75, 100, 150, 200, 300, 400, 500, 600, 700\}$ km (altitude $\geq$ 200 km), with dwell $\leq$ 2200 d and relative rate $\geq 0.01°$/day. After a leg the chaser lies in the target's plane, so its node depends only on (object, epoch). The tour is therefore a *time-dependent orienteering problem*: maximise the number of objects visited subject to a ΔV budget and a mission-time cap, where the arc cost depends on the departure epoch through the evolving gaps. Two envelopes are taken from the draft: "small" (2 km/s, 7 yr) and "large" (5 km/s, 15 yr). The counts include the free start object, since launch is assumed to insert the chaser at the start object. We report distributions over *all* start choices rather than the best of N.

2.4 Solvers and the scalarisation weight

Greedy and beam search rank actions by $w_t\,\Delta t/100\,\mathrm{d} + w_{\Delta V}\,\Delta V/100\,\mathrm{m/s}$. The draft used $w_{\Delta V}{:}w_t = 10{:}1$ and never tuned it. With both budgets binding, we find the Lagrangian-consistent weight by a sweep on random starts and report it on disjoint held-out starts (Section 6). We also test a rollout beam and a permutation NSGA-II (three seeds). The exact reference is a Held–Karp dynamic program over (visited set, current object) with Pareto labels (time, ΔV). It is exact for this leg model up to the weak departure-epoch dependence of drag make-up, and it is checked by brute force on tiny instances (`tests/`).

2.5 Baselines

All baselines use the same pool, leg costs and budgets. The metrics are objects reached, ΔV, time, and propellant mass by the rocket equation.

- *Impulsive*: direct plane-change hops, $2v\sin(\vartheta/2)$ plus Hohmann plus 15 m/s, hop time 2 d, beam width 6.
- *Natural drift only*: no altitude change; the chaser keeps its own orbit while waiting.
- *Published-style drift phasing*: one fixed raised waiting orbit at +40 km (the single raised orbit of [49]), and one fixed waiting orbit at −100 km (the carrier below the targets of [9]).
- *Free per-leg depth, lowering only* (this study).
- *Free per-leg depth, raising or lowering*, with one extra synodic lap allowed (the intermediate drift orbit "below, between or above" of [6]).

2.6 Uncertainty and verification

Sampled legs of the best tour are propagated numerically (Cowell, J_2–J_4, drag, solar radiation pressure, Sun and Moon) with SGP4 targets, and the node closure error is measured. A Monte Carlo then draws closure errors, calibrated on that check, with element-set growth, burn execution, density and lognormal proximity-operations variability, and counts the objects completed within the budget. A one-at-a-time sensitivity study uses paired differences over fixed starts.

2.7 Collision-risk benefit

A kinetic-gas model with latitude-resolved spatial density gives the catastrophic-collision involvement rate of each object against the catalogue population. The cross-section is the sum of radii, a collision is catastrophic if $\frac{1}{2}m_s v^2/m_l \geq 40\,\mathrm{J/g}$, and mass is obtained from the radar cross-section (RCS) by an imputed relation, with 66% of radar cross-sections imputed. Fragment yield follows the standard breakup power law [29]. The model is a ranking and scaling tool with large uncertainty and is *not* a prediction; the draft's superseded "95.4%" claim is not re-derived.

2.8 Key assumptions

The main modelling assumptions are summarised below; their effect is discussed in Section 12.3.

- Legs are impulsive and circular-to-circular, with no in-plane phasing; proximity operations and capture are one fixed allowance of 15 m/s.
- Launch inserts the chaser at the start object at no cost.
- Drag make-up uses a tabulated exponential atmosphere and $C_dA/m = 0.01\,\mathrm{m}^2/\mathrm{kg}$.
- The dry masses and specific impulses of the two propulsion designs, and the kit mass, are stated assumptions rather than sourced values.
- The node uncertainty at epoch (0.01°) is an assumption, since no published RAAN uncertainty was found.

3 Audit of the Draft Study

Table 1 lists the findings that change a headline or a stated claim; the full list (13 analytic and 13 pipeline items) is in `audit/AUDIT.md`. The independent engine reproduces the draft's own tour (29 objects, 1,445.48 m/s, 2,555.54 d, node closure 2×10^{-13} deg at all 28 legs), so the differences between the two studies are model or input choices.

3.1 Reproduction of the 2025 pipeline

An earlier version of this audit could not reproduce the draft's catalogue counts, because the 2025 inputs were not in the repository. With the original files recovered, we ported the two 2025 stages to Python and ran them inside the work folder: the MATLAB cluster locator, which groups non-operational payloads and rocket bodies at 800–1,100 km by a greedy 1° inclination rule, and the Python discovery module (DBSCAN on inclination with $\varepsilon = 0.5°$, then sliding windows of 10–20 objects with altitude, eccentricity and RAAN range limits). Table 2 gives every stage.

Three facts explain the draft's numbers. First, "63,553 rows" is the line count, header included, of the April 2025 catalogue (63,552 objects: 33,505 re-entered and 30,047 in orbit, of which 14,699 payloads, 12,916 debris, 2,368 rocket bodies and 64 unknown). Second, the 1,288 actionable objects come from a different, earlier file, the March 2025 catalogue that the 2025 code actually read, and the rule has no decay filter, so 193 of them had already re-entered; on the April file the same rule gives 1,297. Third, the 40 sub-clusters with 593 objects require the RAAN join of 2025: the sub-cluster windows reject RAAN ranges above 180°, so the result depends on which objects have a node angle. Joining the April 2025 element sets (11,806 sets) and seven cluster exports (389 records, 24 of them with epochs as early as 1984, i.e. the last elements of re-entered objects) gives 662 of the 1,288 objects a RAAN and reproduces the shipped catalogue exactly: the same 593 objects in the same 40 sub-clusters and 8 parent clusters, with 250 of the 270 shipped RAAN values equal to 10^{-4} deg (the other 20 differ by 0.01–0.13°, consistent with averaging over a slightly different set of duplicate element sets). The MATLAB locator output (527 objects in 42 groups) is also reproduced with identical labels. That the DBSCAN step also yields 40 inclination clusters is a coincidence. The numbers in the rest of this paper are unchanged, because no tour uses the sub-cluster catalogue.

The 2025 MATLAB optimiser. The 2025 MATLAB tours differ from the Python results for reasons of model, not of solver quality. The MATLAB optimiser tours one preselected sub-cluster of 20 objects (or one RAAN group of about 25), so its count is capped by the set; it starts at the largest RAAN and admits only targets with a smaller node, with no wrap-around; it ranks

Table 1: Audit Findings That Change a Headline (Severity A) or a Stated Claim (B)

	Draft claim	Finding
A	63,553 rows → 1,288 actionable objects; 40 sub-clusters, 593 objects	Reproduced exactly from the 2025 files (Section 3.1), but the three counts come from two different catalogues and include decayed objects: 193 of the 1,288 and 63 of the 593 had re-entered. On the 2026 snapshot the rule restricted to objects in orbit gives 1,151 and 913 in 51 sub-clusters
A	Hohmann cost symmetric; net advantage $1+4.5x$	$\Delta V_{\downarrow}/\Delta V_{\uparrow} = 1 + \frac{3}{2}x + \frac{9}{8}x^2$; net per-$\Delta V$ gain $1 + 3x$
A	Efficiency has an interior optimum 150–300 km below	No interior optimum (gain per ΔV rises monotonically); quoted values about 11 times too large
A	"59.8% saving" over plane changes	No time-free saving exists. At 1° the saving is 80% if 279 d are allowed, 13% if 30 d. Same-sequence replacement on the draft tour: 72.5%
A	Headline counts (24, 29, 53 objects)	Best-of-60 starts; over all 159 starts the draft's configuration has median 20 (CI 19–20), maximum 28
A	Weights 10:1	Far from the Lagrangian optimum; counts understate the same model by 45–90%
A	"Objects removed"	The tours *visit*; removal needs one kit per object (Section 9)
B	Alignment window 1°	Defined by cost: 12.8 m/s per 0.1°; the 15 m/s allowance covers 0.118°
B	$R > 1$ "for every target"	False for eccentric targets (reversal near $e \approx 0.03$)
B	Greedy and differential evolution	Results come from greedy, beam and NSGA-II; DE exists in the repository but is not behind the numbers
B	Elements	Osculating inputs: nodal-rate noise 0.0026°/day (1σ) against a pool spread of 0.0066°/day

targets with the drift rate of a fixed reference orbit ($a = 7{,}100$ km) and then optimises the per-leg depth (50–400 km) by particle swarm; it uses RAAN values from element sets of mixed epochs without propagation; and it charges no inclination difference, no proximity-operations allowance and no drag. Its SL-8 tour of 20 objects (19 legs, 2,000 m/s, 511 d) and its sun-synchronous tours (26 objects in 4,553 d, or 21 in 1,671 d, each at 2,000 m/s) are therefore not comparable with the 29 (SL-8) and 27 (sun-synchronous) medians of Table 4, although the sun-synchronous figures are of the same order. Several exploratory 2025 outputs carry sign errors (negative or 10^5-day total waits) and are not used. Of the 2025 graphics, the three-panel chart of wait time against forward RAAN difference (exported as a 1.2 s clip) is worth showing as a historical record of the first tours, with the caveat on unpropagated epochs; the 94 s trajectory animation (840×630 pixels) mislabels metres as kilometres and is not.

4 Lowering versus Raising: Exact Results

We first establish the exact comparison between waiting below and waiting above a target. Let $x = \Delta a/a_T \in (0, 1)$ and $f(a) = a^{-7/2}$.

Table 2: Catalogue Stage Counts of the 2025 Pipeline, Reproduced, and the Same Rules on the 2026 Snapshot

Stage	Mar 2025	Apr 2025	Aug 2026
catalogue rows (newest launch)	63,203 (21 Mar)	63,552 (27 Apr)	70,292 (17 Aug)
of which re-entered	33,032	33,505	35,473
numeric apogee, perigee and inclination	62,297	62,642	68,244
mean altitude 0–2,000 km	56,304	56,646	62,272
non-operational rule (R/B, or PAY with status "−")	5,582	5,619	5,736
of which at 600–1,500 km ("actionable")	**1,288**	1,297	1,349
of which re-entered	193	194	198
in orbit	1,095	1,103	1,151
DBSCAN inclination clusters (noise)	40 (13)	39 (14)	40 (19)
sub-clusters / objects, 2025 RAAN joined	**40 / 593**	36 / 552	–
sub-clusters / objects, no RAAN	51 / 900	–	53 / 952
MATLAB locator, 800–1,100 km: objects / groups	527 / 42	530 / 43	–

Rules exactly as in the 2025 code; no decay filter except in the "in orbit" row. The v3 pool of this paper (1,151; 51 sub-clusters, 913 objects) applies the decay filter before clustering. The 2026 sub-cluster counts without RAAN include decayed objects. Source: `analysis/a9_orig2025.py`.

Theorem 1 (Rate gain)**.** *For circular orbits with the same inclination and J_2 only, $R(x) \equiv \dfrac{|\Delta\dot\Omega_\downarrow|}{|\Delta\dot\Omega_\uparrow|} = \dfrac{(1-x)^{-7/2}-1}{1-(1+x)^{-7/2}} > 1$ for every $x \in (0,1)$, independent of i.*

Proof. f is strictly convex, so $f(a-\Delta)+f(a+\Delta) > 2f(a)$, i.e. $f(a-\Delta)-f(a) > f(a)-f(a+\Delta) > 0$. The factor $K\cos i$ cancels. □

Proposition 1 (Series and cost)**.** $R = 1 + \frac{9}{2}x + \frac{81}{8}x^2 + \frac{141}{8}x^3 + O(x^4)$*, with* $0 \le R - 1 - \frac{9}{2}x \le 13.58\,x^2$ *for* $x \le 0.15$ *(supremum checked on a 20,000-point grid and symbolically). The exact Hohmann costs satisfy* $\Delta V_\downarrow = \frac{1}{2}x + \frac{3}{8}x^2 + \dots$ *and* $\Delta V_\uparrow = \frac{1}{2}x - \frac{3}{8}x^2 + \dots$ *(units of v_c), hence* $\Delta V_\downarrow/\Delta V_\uparrow = 1 + \frac{3}{2}x + \frac{9}{8}x^2 + O(x^3)$*, and the gain per unit ΔV is larger for lowering by* $R\,\Delta V_\uparrow/\Delta V_\downarrow = 1 + 3x + O(x^2)$.

At $a_T = 7350$ km (975 km altitude), $R = 1.031, 1.063, 1.130$ and the equal-ΔV gain ratio is $1.021, 1.042, 1.087$ for offsets of 50, 100 and 200 km (first-order $1+3x$: 1.020, 1.041, 1.082). The error order $O(x^3)$ stated in the draft is therefore wrong, since the remainder is $O(x^2)$. Figure 1 shows these results.

Both signs, inclination and eccentricity. For retrograde orbits ($\cos i < 0$) the rates change sign and "lowering" accelerates the drift eastward; Theorem 1 is unchanged. Including J_2^2 and J_4 changes R by at most 1.1×10^{-3} across $i = 52$–$98.7°$. A circular chaser against an *eccentric target* sees an effective altitude offset, because $\dot\Omega_t \propto a^{-7/2}(1-e^2)^{-2}$ is the rate of a circular orbit lowered by $\frac{4}{7}e^2 a$ (0.1 km at $e = 0.005$, 10.5 km at 0.05, 42 km at 0.1). For a 100 km offset, $R = 1.063, 1.054, 1.028, 0.861, 0.433$ at $e = 0, 0.01, 0.02, 0.05, 0.1$, so raising wins above $e \approx 0.03$. This is irrelevant to the SL-8 pool ($e \approx 0.002$) but affects 62 of the 593 shipped objects.

Drag. The $1+\frac{3}{2}x$ cost ratio applies to the transfer alone. When the dwell make-up is added, the lower-to-raise ΔV ratio at 975 km is 1.009 (50 km offset, mean activity); near 600 km it is 1.1 (mean) to 1.4–1.5 (high activity). The draft's "2 km/day drag floor" never binds (decay at 300 km altitude is 0.36 km/day); the effective limits are the 200 km altitude floor and the make-up cost (about 1 m/s per day at 275 km).

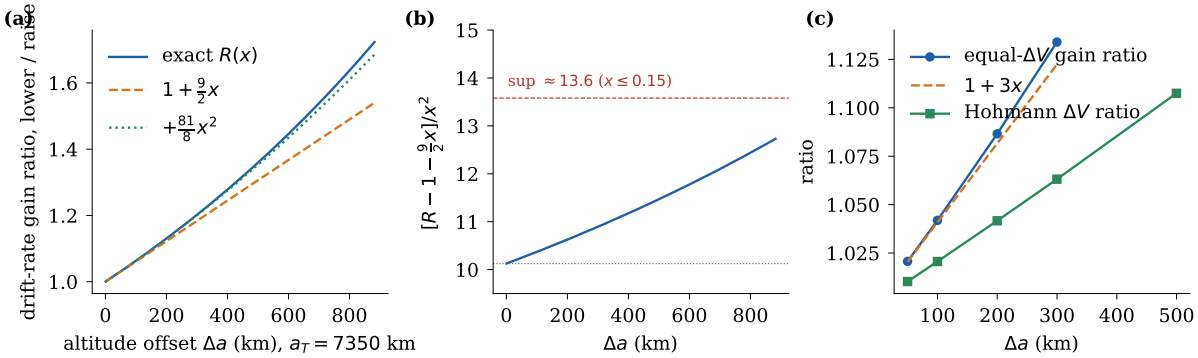


Figure 1: Lowering versus raising for circular orbits. (a) Exact rate-gain ratio $R(x)$ and its series. (b) Scaled remainder, bounded by 13.58. (c) Hohmann cost ratio and equal-ΔV gain ratio against $1 + 3x$. Lowering wins in both rate and cost, but by only a few per cent at practical offsets.

Decision rule. Lowering only closes gaps of one sign (westward for prograde orbits). For a gap $g \in (0, 360°)$ in the lowering direction, lowering is faster when $g < g^* = 360R/(1 + R) \approx 180°(1 + \frac{9}{4}x)$; the approximation errs by $\le 0.46°$ for $x \le 0.1$. Allowing both senses halves the expected wait for a uniform random gap (Monte Carlo ratio 1.94, theory 2), and on SL-8 pairs the rule picks the faster sense in 94–96% of cases (wait ratio 2.15 at 50 km). The "always lower" design rule is therefore a weakness rather than a result, and the bidirectional planner is the right baseline.

5 Validation of the Numerical Model

Secular rate. Against a J_2-only Cowell integration, Eq. (1) evaluated on mean elements is within 0.04–0.08%; with the osculating a it is 0.34–0.46% off. Adding J_3 and J_4 shifts the rate by 0.02–0.23%, an effect dominated by J_4 (Fig. 2).

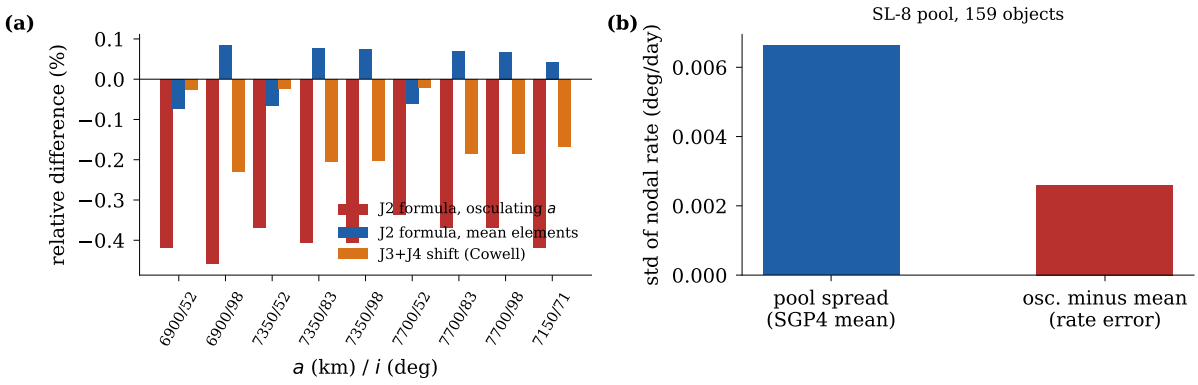


Figure 2: Validation of the secular nodal rate. (a) Secular-formula error against Cowell integration for nine (a, i) cases. (b) Pool spread of the nodal rate against the rate error from using osculating elements; the osculating error is a significant fraction of the spread, which is why mean elements are used.

Leg closure. For eight legs of the tuned best tour, a circular Cowell chaser (full forces) and SGP4 targets give node closure errors of $-0.002 \pm 0.017°$ (mean $\pm$ s.d., maximum 0.039°) for the first-order plan and $-0.006 \pm 0.018°$ (maximum 0.049°) for the second-order plan. These correspond to about 1.3–1.5 m/s of correction per leg, against the 15 m/s allowed (0.118°). A J_2-only numerical model disagrees by 0.08° because the plan includes J_4; adding J_3 and J_4 closes this

to 0.003°, and drag, SRP and Sun/Moon account for the rest (Fig. 3). Planned and propagated dwell differ by up to 3 d. Eight legs is a small sample, and Section 12.3 states what is not covered.

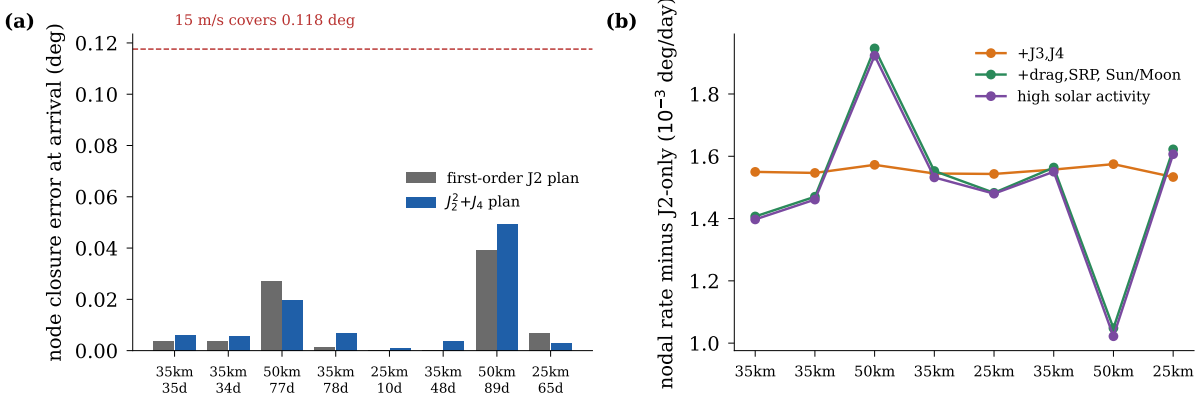


Figure 3: High-fidelity verification of the leg model. (a) Node closure error at arrival on eight sampled legs; the dashed line is the error covered by the 15 m/s allowance. (b) Nodal-rate increment of each added force model over J_2 only.

Code. Twenty automated tests pass. They cover the zonal accelerations against the potential gradient; the series and bound of Proposition 1; the Hohmann ratio and the plane-change small-angle limit; the second-order rate against SGP4; the drag identity; the replay of the draft's tour; the closure and budget invariants; the dynamic program against brute force; the funnel counts; the risk-model scaling; the SMAD learning-curve, pass-duration and cost-uncertainty relations; and the exact reproduction of the 2025 counts.

6 Optimiser Validation

Weight. We swept $w_{\Delta V}$:w_t on random starts and evaluated the result on disjoint held-out starts. The best weights are 2 (small envelope) and 1.5 (large), against 10 in the draft, and the held-out medians are 31 versus 22 objects (small) and 68.5 versus 39.5 (large). Figure 4 shows the sweep. This is standard multiplier tuning, not a new technique; the point is that the draft's published counts understate its own model.

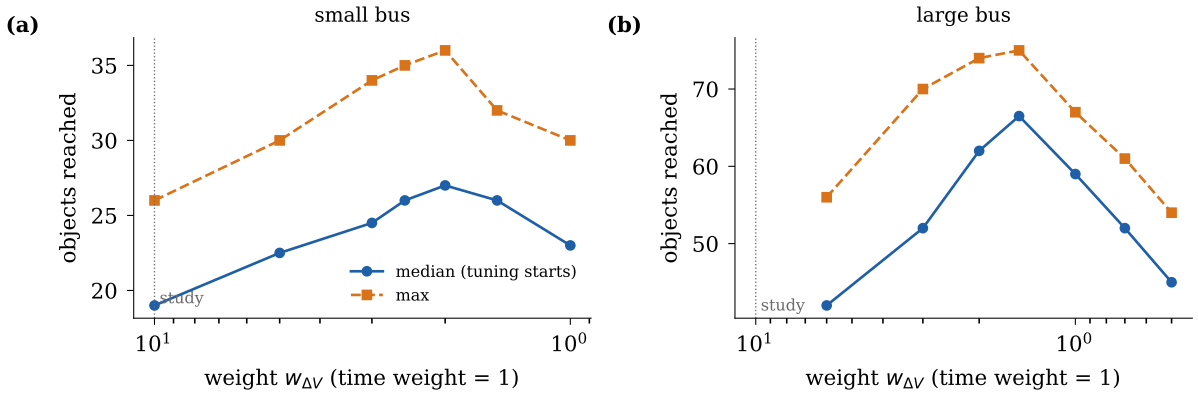


Figure 4: Objects reached against the scalarisation weight. The draft used 10, well away from the best values for either envelope.

Distance from optimum. On 32 small instances (16 windows of 10 and of 12 consecutive-RAAN objects, with budgets of 700 m/s and 600 d, and of 800 m/s and 700 d) the exact dynamic program gives the optimum count. The mean gaps (95% bootstrap CI) are: greedy at the draft weight 18.7% (16.0–21.4); beam-20 at the draft weight 10.3% (8.0–12.7); NSGA-II 9.7% (7.1–12.2); tuned greedy 6.1% (4.1–8.2); tuned beam-20 1.9% (0.6–3.4), optimal in 81% of instances (worst 12.5%); and rollout beam 2.6% (1.0–4.2). Beam width saturates at about 10 on the full pool (best start 34, 34, 35, 35, 35, 35, 35 objects for widths 1–1000; Fig. 5). The ΔV-only relaxation bound is on average 31% above the exact optimum on the small instances. On the full pool it is 71 objects at 2 km/s (best found 36) and vacuous at 5 km/s (160 against a pool of 159), so the full-pool gap is not certified and the small-instance gaps are the evidence. The draft's belief that wider beams or evolutionary search would help is not supported; the weight mattered more.

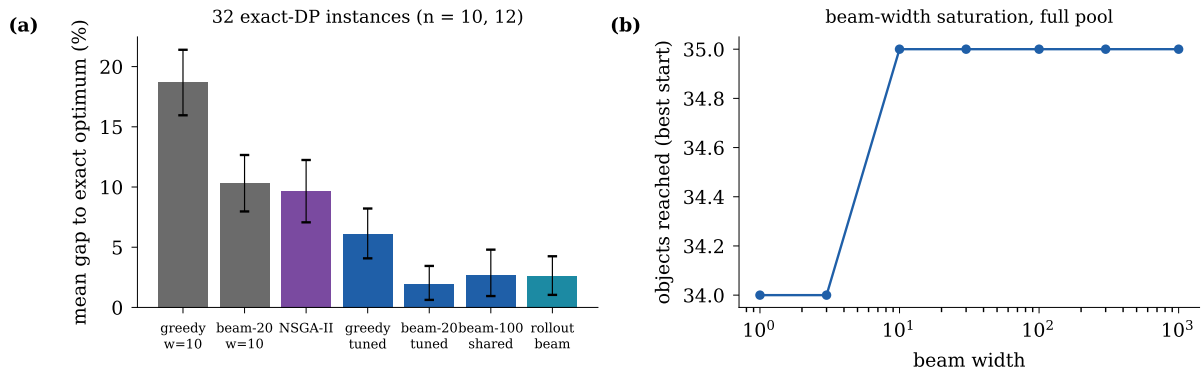


Figure 5: Optimiser performance. (a) Mean gap to the exact optimum with 95% CIs. (b) Beam-width saturation on the full pool. The tuned beam search is close to optimal, and widening the beam beyond about 10 gives no gain.

7 Results

7.1 Fair comparison with natural drift and impulsive plane changes

To place the drift tours in context, we compare them with every baseline at equal budget. Table 3 gives the medians over all 159 starts (bootstrap CI), with ΔV, time and propellant, and Figure 6 shows the full distributions. Three comparisons from Table 3 warrant discussion. First, at equal ΔV budget the drift tour reaches 3.2 times (small) and 3.5 times (large) as many objects as impulsive hopping, but it needs 2,500–5,300 d against 16–36 d. The saving is a trade of time for propellant, and the value of that trade depends on the bus life, not on a universal percentage. Second, waiting with no altitude change is cheap (114 m/s) but slow, reaching only 6 objects. Third, the published-style *fixed* waiting orbits do worse than free per-leg depth: the +40 km orbit reaches 16 (median) and the fixed −100 km orbit 18, against 29. The −100 km result is deterministic here, since every leg costs nearly the same and the budget binds. The gain over the published rules therefore comes from the per-leg choice of depth and sign. The bidirectional planner adds about two objects at 2 km/s and nothing measurable at 5 km/s.

7.2 Headline missions on the frozen snapshot

Table 4 gives the headline re-run of the two envelopes with the tuned weights. The draft's 29 is reproduced exactly by the independent engine with the draft's settings, and it is the maximum over starts of a configuration whose median is 20. With the weight corrected, the median is 29

Table 3: Objects Reached (Including Start), SL-8 Pool, All 159 Starts

Method	median	95% CI	max	ΔV (m/s)	time (d)	chem (kg)	EP (kg)
small envelope: 2 km/s, 7 yr							
draft as audited	20	19–20	28	1664	2555	489	89
v3 inputs, draft weight	20	19–22	27	1707	2554	506	91
fixed raise +40 km	16	14–17	30	851	2451	218	44
fixed lower −100 km	18	18–18	18	1977	1010	614	107
free lower, tuned	29	27–31	36	1968	2513	611	106
free bidirectional + laps	31	28–32	36	1971	2479	612	106
natural drift only	6	6–7	12	114	2267	26	6
impulsive plane change	9	8–10	15	1775	16	532	95
large envelope: 5 km/s, 15 yr							
draft as audited	35	32–38	52	2500	5477	853	137
fixed raise +40 km	32	29–36	51	1779	5329	534	95
fixed lower −100 km	44	44–44	44	4998	2732	2742	295
free lower, tuned	67	64–68	76	4976	5325	2718	293
free bidirectional + laps	67	67–70	77	4974	5342	2716	293
natural drift only	12	12–13	22	440	5138	105	23
impulsive plane change	19	18–22	31	4779	36	2510	280

Propellant (fixed-orbit rows: rocket equation on the median ΔV): storable chemical design (700 kg dry) and electric design (900 kg dry, 1800 s), rocket equation.

Table 4: Headline Re-run on the Frozen Snapshot

	2 km/s, 7 yr	5 km/s, 15 yr
tuned weight	2	1.5
median over all starts (95% CI)	29 (27–30)	67 (64–68)
greedy median	28	65
best start	36 (1995 m/s, 2397 d)	76 (4981 m/s, 5090 d)
best-of-60 starts, mean (CI)	35.3 (34–36)	75.9 (75–76)
Monte Carlo, best tour: refreshed	33.0	71.8
unrefreshed (open loop)	28.7	58.9
pessimistic errors	30.3	67.4
draft as published (best of 60)	29 (1445.5 m/s, 2555.5 d)	53 (4101 m/s)
four bands (SL-8, SSO, Iridium, SL-16), medians summed	87 (best 101)	171 (best 188)

SL-8 pool, beam width 12 on all 159 starts unless stated; Monte Carlo of execution errors on the best tour.

and the best 36. In the large envelope the pool is nearly exhausted by the best tours (76 of 159), so the counts there are pool-limited and are not a measure of the method's reach. Figure 7 shows the two best tours.

7.3 Collision-risk benefit

To connect the tours to the debris environment, we compute the catastrophic-collision involvement rate removed by each median tour (Table 5, Fig. 8). For the SL-8 median tour the removed hazard is 18% of the pool's total (0.0088 events/yr). Over 25 years the model gives 0.040 catastrophic collisions avoided (Monte Carlo 5–95%: 0.025–0.055) and about 65 fragments > 10 cm (21–217), while a random set of the same size gives 0.039. In SL-8 the per-object hazard is nearly uniform (CV 0.16), so *risk-aware selection is worth nothing* there. In the sun-synchronous band it would double the benefit (CV 0.67), but the tour as sequenced captures only half of what the top-n set would. Choosing the start by risk raises SL-8 from 28 to 36 objects and the benefit from 0.040 to 0.050. The cost per avoided event is a scaling statement, not an estimate: at an assumed \$100–400 M per mission, the nominal figure is \$2.5–10 B per avoided catastrophic collision (\$1.8–16 B

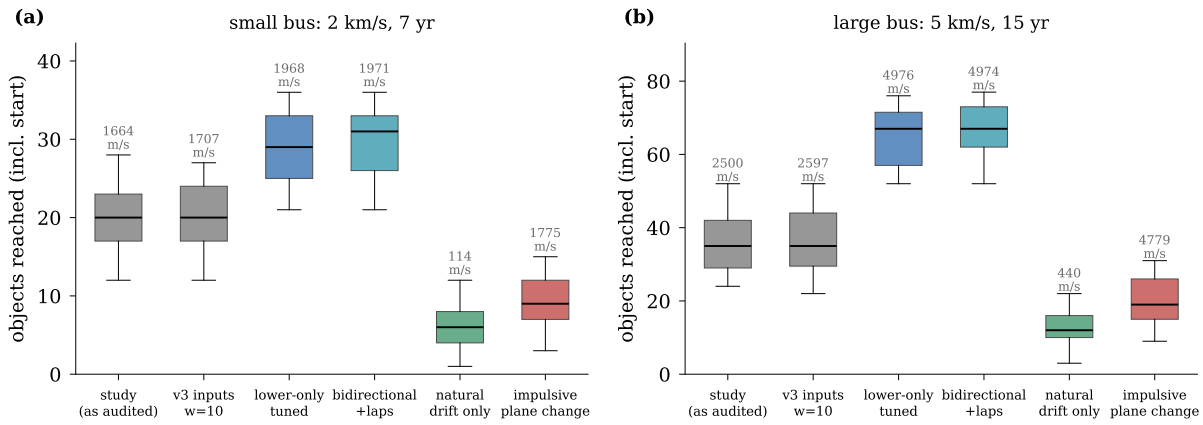


Figure 6: Objects reached over all 159 starts for each method (box: quartiles; whiskers: range); labels give the median ΔV. Free per-leg depth reaches the most objects, while impulsive hopping is the only fast option.

Table 5: Catastrophic-Collision Involvement Rate Removed by the Median Tour (Kinetic-Gas Model; *Ranking-Level Accuracy Only*)

band	pool	tour n	tour rate (10^{-3}/yr)	top-n	random	CV
SL-8	159	29	1.58	1.85	1.61	0.16
SSO	135	27	1.27	2.69	1.31	0.67
Iridium-class	28	11	0.55	0.65	0.53	0.27
SL-16	20	16	0.87	0.89	0.87	0.10

Tour n is the median object count including the free start; the benefit calculation below uses the median-hazard tour (28 objects for SL-8). CV: coefficient of variation of per-object rate; "top-n": removing the n highest-rate objects; "random": random n.

across the 5–95% interval of avoided events). Published cost anchors are sparse and are not used as inputs. The result must not be read as a case for or against removal, because the model has no fragment-cascade feedback beyond the first generation (amplification 1.06) and the long-term benefit of avoiding the *first* fragmentation in a band is not captured.

8 Uncertainty and Sensitivity

Execution errors. The closure-error Monte Carlo (calibrated on Section 5) yields, for the planned 35-object tour at the 2 km/s budget, 32.0 objects completed with an element refresh at each leg (a receding-horizon replan), 27.8 without refresh (the 95th percentile node error is then 0.36°, three times the 15 m/s allowance) and 28.8 under pessimistic errors. A planning margin of 0–15% of the budget does not help the open-loop case (27.5–27.9 at every margin), so the recovery comes from refreshing the elements and not from reserve (Fig. 9). The first-order draft model and the second-order model give nearly the same count (32.1 and 32.0), because the rate bias is common mode.

One-at-a-time sensitivity. The sensitivity study uses paired differences over fixed starts (baseline 28.1 objects, Fig. 10). The largest losses come from a relative-rate floor of 0.03°/day (−9.8, CI −11.1 to −8.5); the draft weight of 10 (−8.9); a proximity-operations allowance of 50 m/s (−8.2) or 30 m/s (−3.7); an in-plane phasing allowance of 30 m/s and 20 d per leg (−7.3); a 5 yr life (−5.6); and a 1.5 km/s budget (−5.5). The gains come from a 3 km/s budget (+2.6), a 10 yr life (+1.8) and a 5 m/s allowance (+1.6). The ballistic parameter (0.003–0.03), solar

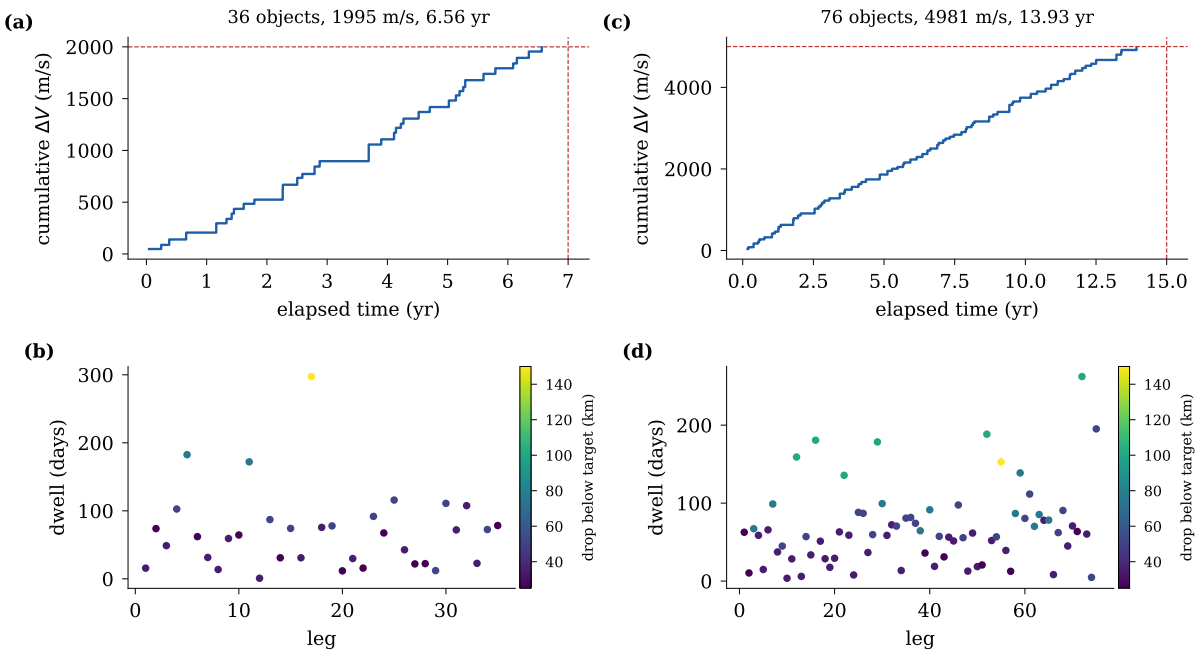


Figure 7: Best tours for the two envelopes. (a,c) Cumulative ΔV against time; (b,d) dwell per leg, coloured by depth below the target.

activity, the decay cap, an altitude floor of 200 km, the leg-wait cap and inclination charging have exactly zero effect. The drag insensitivity is a property of this pool (drops of 25–150 km below 975 km, where decay is negligible) rather than a general result, and it would not hold for bands near 600 km. In-plane phasing is not otherwise modelled (closing 90–180° over 20 d costs 4.5–9 m/s round trip, and 18–36 m/s over 5 d); this is the most important unmodelled cost.

Beyond secular J_2. At 83° the absolute luni-solar nodal rate is 6×10^{-5}°/day, and its *differential* across a 100 km drop is 1.2×10^{-6}°/day, or 3×10^{-5} of the J_2 differential. With drag, SRP and third bodies, the full Cowell model differs from J_2–J_4 by 1×10^{-4}–4×10^{-4}°/day (below 0.05%). The SGP4 element-set error is the dominant data uncertainty: the TLE range error grows by about 1.5 km/day [36], no published RAAN uncertainty was found, and the node uncertainty at epoch (0.01°) is an assumption.

9 Novel Methods

Each idea is simulated against the baselines of Table 3 on the SL-8 pool with fixed seeds; the objective is the number of objects reached or placed, ΔV, time, and launch or propellant mass. The verdicts use only the numbers, and "prior art" cites what Section 12.2 found. Table 6 summarises the screen and Figure 11 shows the carrier comparison.

The one supported idea. A carrier parks on a *single persistent* lowered drift orbit (depth 400 km, $\Delta V \approx 210$ m/s including make-up) whose plane sweeps through the cluster. When the node of its orbit coincides with that of a target within the tolerance, a kit departs (kit ΔV 260 m/s, 5 days, 30 m/s extra), and the carrier does not return to any target plane. The K nearest-in-time targets are served in the order in which the sweep reaches them. Depth trades time against launch mass (Fig. 11), and the time CI is a bootstrap over all 159 starts. The comparison fixes the kit mass at 135 kg (unverified), counts kit propellant with $I_{sp} = 280$ s, and omits kit avionics,

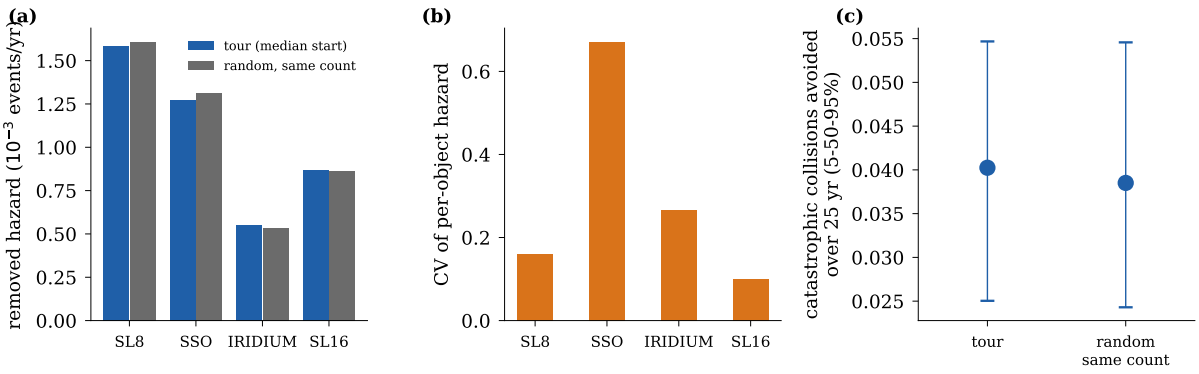


Figure 8: Collision-risk benefit of the median tours. (a) Removed hazard of the median tour and of a random set of equal size. (b) Heterogeneity of per-object hazard. (c) Collisions avoided over 25 yr. Ranking-level accuracy only.

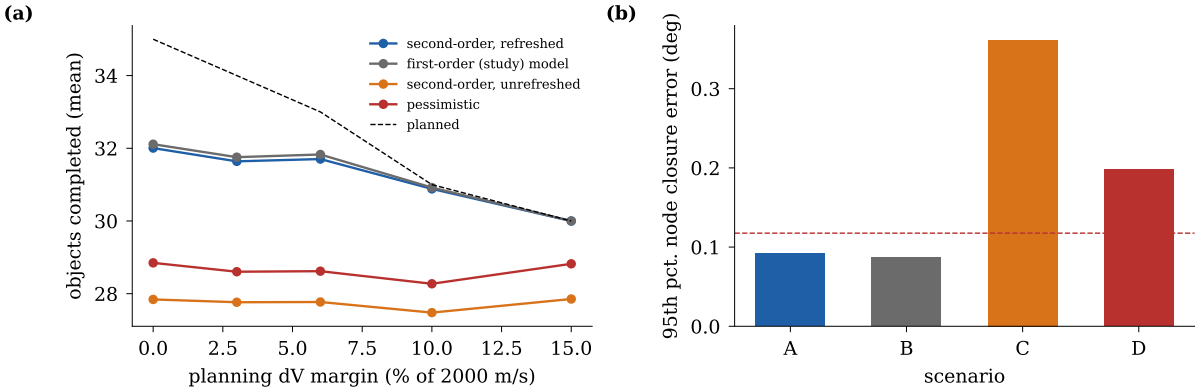


Figure 9: Monte Carlo of execution errors. (a) Objects completed against planning margin for four scenarios. (b) 95th percentile closure error (A: second-order refreshed; B: first-order draft; C: unrefreshed; D: pessimistic; dashed: error covered by the allowance). Refreshing the elements, not adding margin, recovers the lost objects.

capture hardware and rendezvous autonomy. The launch-mass advantage (5% at $K = 8$, 20% at $K = 16$, 50% at $K = 36$) is therefore an upper bound. The time advantage (4.5 to 6.3 times) is less sensitive: kit allowances of 60–100 m/s and 10–20 d change the median at $K = 16$ from 298 to 313 d at 300 km depth. The hopper medians use 20 sampled starts. Prior art covers a carrier with kits that waits in a drift orbit [8, 9, 49]; whether kits depart from a persistent orbit without a carrier return was not found in the sources reached, but absence in a bounded search is not proof.

10 Servicer Design and Comparison with Demonstrations

The tours above count visits by a point mass. This section asks what spacecraft would fly them, following the subsystem-level process of SMAD [35]. Unless stated, the reference is the supported architecture of Section 9: a chemical carrier parked 400 km below the SL-8 cluster that releases $K = 16$ kits (the median start, Table 6), against a serial hopper that carries the same kits, and a visit-only hopper on the 2 km/s, 7 yr envelope. Inputs that are assumptions are tagged (A) in Appendix B; the computations are in `analysis/a8_servicer.py`.

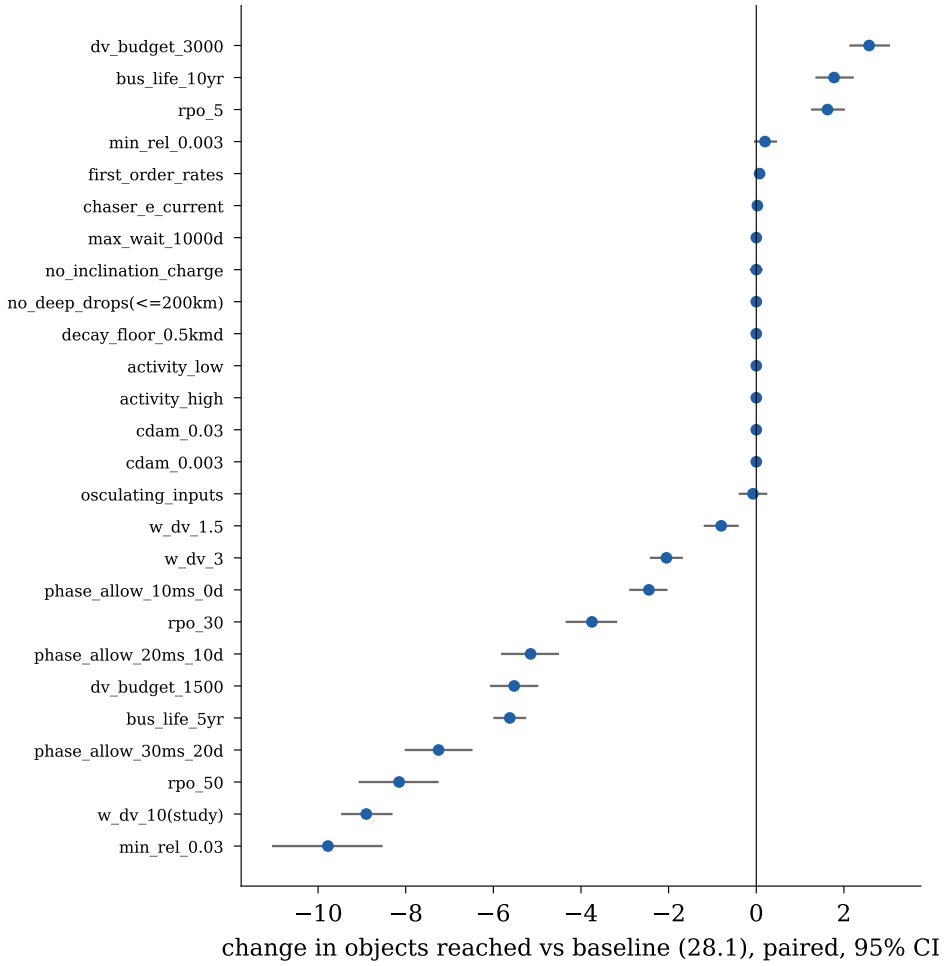


Figure 10: One-at-a-time sensitivity: change in objects reached against the baseline, paired, with 95% CI. The proximity and phasing allowances dominate; drag has no effect for this pool.

10.1 Where the drift-based tour fits

We first place the concept against the servicing and removal missions that have flown or are funded (Table 7, Fig. 12). Masses and costs are as published by the agencies or in the NASA servicing survey [3]; values found only in databases or trade reports are marked, and no mission in the sources reached publishes its ΔV capability.

Table 7 shows a clear pattern. The demonstrations retire the risk of one step at a time: capture of a prepared or own client (RemoveDEBRIS, ELSA-d), non-cooperative rendezvous with a rocket body to about 15 m (ADRAS-J, reported in a secondary source), and capture of one unprepared object (ClearSpace-1, ADRAS-J2). In the terms of SMAD Table 20-15 [35, p. 804], rendezvous with a non-cooperative rocket body has been tested in space, while capture of a tumbling, unprepared rocket body has not, so it sits at about TRL 4–5 (our reading). The drift-phased tour does not compete with these missions; it is the step after them. Its advantage is the number of objects per launch, which no demonstration addresses, and its cost is the years of drift and the need for many kits that each perform a capture. The GEO life-extension servicers, which move between clients, and the pods installed by a robotic vehicle on several clients [3] are the closest operational analogue of a carrier that distributes kits.

Table 6: Screen of Ten Ideas plus the High-Fidelity Check (EP: Electric Propulsion)

Idea	Result	Cost	Verdict
Continuous low-thrust profiles	Objects 25.5–28 (CI 24–31) against 28 impulsive; propellant 97–105 kg against 610 kg, entirely from I_{sp} (1800 s against 320 s)	thruster, power, time	no trajectory gain; prior art [6, 30]
Eccentric drift orbit / apsidal	Gain per ΔV 0.970–0.998 of circular, falling with depth	none	rejected
Oberth-type plane change at high apogee	Direct plane change cheaper than the bielliptic option for gaps up to 30° (127.5 against 163 m/s at 1°); drift 25–55 m/s for 1–10° in 2 yr	–	rejected for small angles
Luni-solar plus J_2	Differential rate 3×10^{-5} of J_2	–	rejected
Drag sail as altitude shaper	Time to descend 100 km for C_dA/m = 1–0.1 m^2/kg, mean activity: 16–156 d at 500 km, 77–767 d at 600 km, 330–3,270 d at 700 km, 840–8,370 d at 775 km, 4,700–47,000 d at 975 km (high activity 5–9 times faster)	sail area and mass; no control of re-raise	useful below about 700 km; impractical at 975 km
Deorbit of removed object to retime	No mechanism: the removed object leaves the pool, the chaser's orbit is unchanged. Towing a 1,435 kg object 100 km down costs 50.6 m/s, triples the propelled mass, and does not deorbit it at 975 km	–	rejected
Better tour formulations	Tuned weight: +9 objects (small), +32 (large) median; rollout beam and exact-gap results (Section 6); receding-horizon refresh +4.2 objects (Section 8)	compute only	supported, but standard
Two or three servicers, one cluster	Same total ΔV split 1/2/3 ways: 35/36/39 objects, time 2493/1380/887 d, propellant 621/513/477 kg; k full servicers: 35/65/94	more hardware	time and mass benefit, no count benefit
Band choice by RAAN concentration	40-object pools: RAAN-window 21, random 16, evenly spread 14 objects	–	supported (pool design)
Parked carrier + expendable kits	K = 16: placement 219 d (CI 195–246) against 1198 d for a hopper; launch mass 3.3 t against 4.1 t; kit $\Delta V \leq 261$ m/s; K = 36: 484 d against 3041 d, 6.5 t against 13.4 t	kit autonomy and kit bus mass (not counted)	**candidate**; partial prior art
(verification) Full Cowell model	Full Cowell against J_2–J_4: slope differences $\leq 4 \times 10^{-4}$°/day	–	supports the model

10.2 Trade space

To show how objects per mission scale with the two budgets, we re-ran the tuned beam search over a grid of ΔV budgets (1–5 km/s) and mission lives (3–15 yr) on 40 fixed starts. Figure 13 shows the medians. The count is limited by whichever budget binds: with 3 yr it stalls at 14–17 objects whatever the ΔV, and with 1 km/s it stalls at 14–15 whatever the time. At 7 yr the count rises from 15 (1 km/s) to 28 (2 km/s) and only to 33 (4–5 km/s), so above about 2 km/s the chemical propellant grows faster than the count. Long lives pay only with large budgets: 15 yr and 5 km/s give 63 objects. A visit-only servicer is therefore best sized at about 2 km/s for a 7 yr life, which is the small envelope of the draft.

10.3 Propulsion: chemical or electric for the drop-and-return profile

Each leg drops 25–700 km, waits, and returns, so the propulsion choice sets both the propellant and the time spent manoeuvring. A storable bipropellant gives I_{sp} = 300–340 s and a Hall thruster 1,500–2,500 s [35, Table 17-4, p. 692]; we keep the stated 320 s and 1,800 s. For the best small-envelope tour (36 objects, 1,995 m/s, 2,397 d), the electric servicer needs 108 kg of propellant against 622 kg, and its launch mass falls from 1,322 to 1,008 kg despite the heavier stated dry mass (900 against 700 kg). The price is thrust time. With one 1.35 kW, 83 mN thruster [35, Table

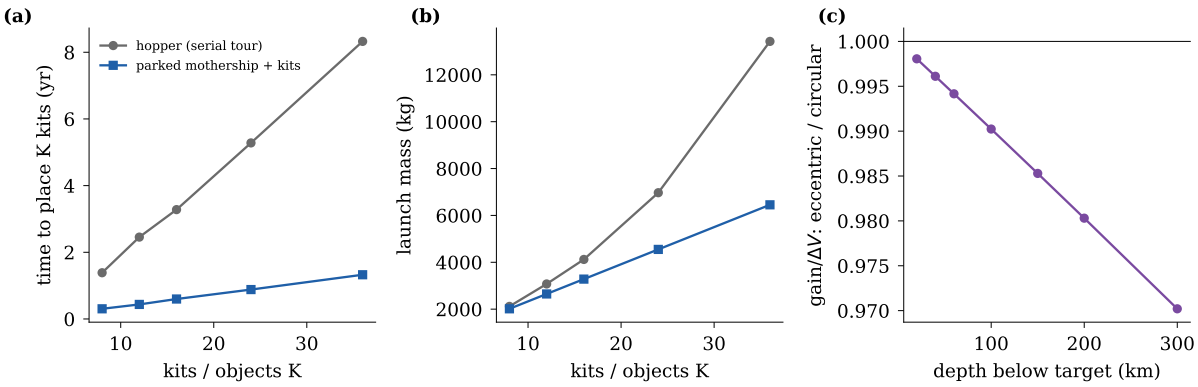


Figure 11: Novel-method comparisons. (a) Time to place K kits: serial hopper against a parked carrier at 400 km depth. (b) Launch mass. (c) Eccentric against circular drift orbit, gain per ΔV. The carrier places kits several times faster; the eccentric drift orbit gives no gain.

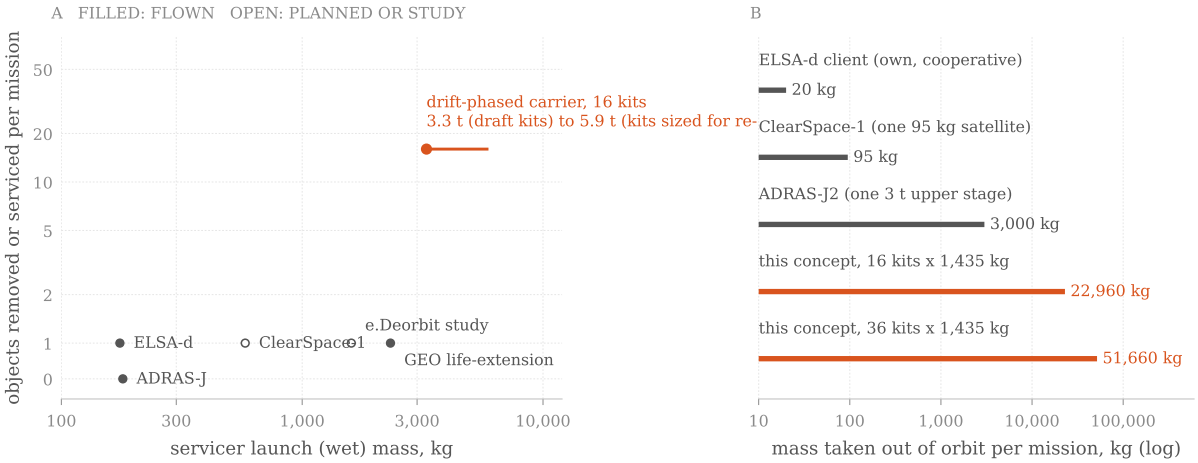


Figure 12: Servicer launch mass against objects per mission, and mass taken out of orbit per mission. Every flown or planned demonstration handles at most one object or its own test targets, while a 16-kit carrier would remove about 23 t of rocket bodies. Filled markers: flown; open: planned or studied; sources in Table 7.

17-11, p. 707] firing in sunlight only, the transfers take 6,370 thruster hours, 17% of the mission, against a demonstrated life of more than 5,700 h for that class [35, p. 707]; the large envelope needs 17,450 h, or about three thruster lifetimes. The continuous-thrust tours of Table 6 reached 25.5–28 objects against 28, so the spirals cost about one object in this model. For the carrier the answer reverses. Its only large manoeuvre is the 209 m/s descent, the electric version saves 173 kg of propellant but is heavier at launch (3,309 against 3,282 kg), and the spiral down takes about 153 d, which delays every kit (Fig. 15). We therefore recommend chemical propulsion for the carrier and its kits, and electric propulsion only for a visit-only inspection tour, with at least two thrusters.

10.4 Capture, guidance and relative navigation

A rocket body offers no docking fixture, may tumble, and has a known geometry (nozzle, payload-adapter ring, cylindrical tank). The options demonstrated or planned are a net and a harpoon (RemoveDEBRIS), a magnetic plate on a prepared client (ELSA-d), robotic arms with blockers (ClearSpace-1) and an arm grasping the payload attach fitting (ADRAS-J2, secondary source)

Table 7: Servicing and Removal Demonstrations Compared with the Drift-Phased Carrier

Mission (agency)	Year	Wet mass	Target	Capture	Objects	Cost, published
RemoveDEBRIS (EU) [3, 13]	2018, flown	n/f	own CubeSat targets	net, harpoon	2 own	EUR 15.3 M total (EU 7.0 M)
ELSA-d [3]	2021, flown	175 kg†	own client, ≤20 kg, prepared	magnetic plate	1 own	n.p.
ADRAS-J (JAXA) [27, 28]	2024, flown	180 kg	H-IIA upper stage, about 3 t	none (inspection)	0	n.p.
ADRAS-J2 (JAXA) [27]	FY2027, planned	n/f	same upper stage	robotic arm†	1	JPY 13.2 bn†
ClearSpace-1 (ESA) [14–16]	2029, planned	580 kg	PROBA-1, 95 kg, unprepared	capture and blocker arms	1	about EUR 100 M (ESA 86 M)
e.Deorbit (ESA study) [17]	to 2018	1,600 kg†	one large ESA derelict	arms, net or harpoon studied	1	n.p.
UK national ADR (UKSA) [50]	about 2029†	n/f	two UK-licensed defunct satellites	n/f	2	about GBP 61 M†
GEO life-extension servicer [3]	2020, operational	2,326 kg†	cooperative GEO client	docking	1 at a time	n.p.
This study: carrier with kits	concept	3.3–5.9 t	16 SL-8 bodies, 1,435 kg each (A)	one kit per object	16 (8–36)	566 M USD (2024, estimate)

† secondary source (database, trade press or filing; see `research/missions_v4.json`). n/f: not found in the sources reached; n.p.: not published. Commercial operators are not named.

[3, 15]. For kits on tumbling bodies, the option that needs the least synchronisation is a contact attachment to the adapter ring or nozzle after the kit has matched the body's motion, or a harpoon; a net suits small objects but not a 1.4 t body that must then be pushed. We do not size the capture mechanism; this is the largest open item of the design. The navigation sequence follows the demonstrations: angles-only far-range tracking with a visible camera, mid-range lidar and thermal imaging, and close-range pose estimation from 3D lidar, the kind of vision-based navigation that RemoveDEBRIS tested [3]. These sensors, the kit interfaces and the capture hardware form the "payload" of the servicer in the budgets below.

10.5 Communications and operations

The drift waits are long and quiet, but the approaches are short and critical. At 975 km the longest pass over a station with a 10° mask is $T_{max} = P\lambda_{max}/180° = 12.4$ min [35, Eq. (5-52), p. 121], where P is the period and λ_{max} the maximum Earth central angle, and 8.2 min at the 573 km parking orbit. A numerical contact simulation for the 82.9° orbit gives 4.5 passes and 43 min per day for a station at 40° latitude, with gaps of up to 11.5 h, against 13.8 passes, 143 min and gaps of at most 1.6 h for a station at 78°. Even with a high-latitude station the servicer is in contact for less than 10% of the time, so the final approach, capture and kit release must be autonomous, with the ground approving each phase (go/no-go) rather than flying it. During the drift the operations are light: an element-set refresh and a small trim per leg, which Section 8 showed recovers four objects over open-loop execution. A two-station network, one at high latitude, and a small team are enough; we assume 12 staff-years per year (A) for the cost estimate.

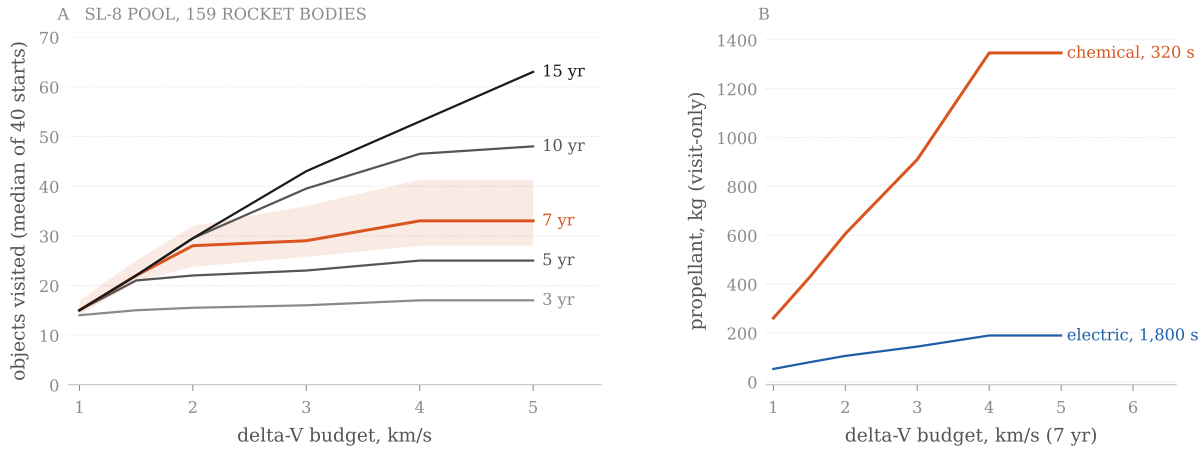


Figure 13: Objects visited against ΔV budget and mission life (median of 40 starts; band: interquartile range at 7 yr), and propellant for the 7 yr case. Both budgets must grow together. SL-8 pool, frozen snapshot, beam width 12, tuned weights.

10.6 Power and mass budgets

To size the power system we prepared an operating budget for three modes (A): drift cruise, proximity operations with kit release, and electric thrust. The loads, with a 25% margin for a new design [35, Table 10-9, p. 316], are 362 W, 775 W and 1,712 W. The array follows

$$P_{sa} = \frac{P_e T_e / X_e + P_d T_d / X_d}{T_d}, \qquad P_{EOL} = P_0 I_d \cos\theta \, L_d, \tag{2}$$

[35, Eqs. (11-5)–(11-8), pp. 413–417], where P_e and P_d are the loads in eclipse and daylight, T_e and T_d the eclipse (35.0 min at $\beta = 0$) and daylight durations, $X_e = 0.60$ and $X_d = 0.80$ the path efficiencies for peak-power tracking, $P_0 = 301\,\mathrm{W/m^2}$ for 22% multijunction cells, $I_d = 0.77$, $\theta = 23.5°$ and $L_d = (1 - 0.005)^{\mathrm{life}}$ [35, Tables 11-34 to 11-36, pp. 412–414]. The battery capacity is $C_r = P_e T_e / (\mathrm{DOD}\, n)$ [35, Table 11-40, p. 422] with a depth of discharge of 25%, the upper end of the 15–25% that LEO cycling allows [35, p. 420]. For a chemical servicer the array is 7.9 $\mathrm{m^2}$ and 65 kg (25 W/kg, [35, Eq. (10-13), p. 333]) and the battery 2.0 kW h and 22 kg at 90 W h/kg; for the electric servicer the array grows to 11.9 $\mathrm{m^2}$ and 98 kg if the thruster fires in sunlight only. Table 8 gives the mass allocation, obtained by applying the average subsystem fractions of SMAD Appendix A [35, Table A-2, p. 895] to the stated dry masses, and Figure 14 the launch masses. The bottom-up array and battery (87–108 kg) are well below the 224 kg allocated to power, so the stated 700 kg dry mass leaves room for the capture hardware; the fractions come from communication satellites and are a check, not a design.

10.7 De-orbit kits

The draft fixed the kit at 135 kg and charged it only the rendezvous ΔV (about 254 m/s, 13 kg of propellant per kit). A kit must also lower the rocket body. We computed the perigee that gives a residual life of 25 or 5 years with an orbit-averaged drag model (same atmosphere as Section 2.1, density scaled by 0.5 and 2), using the median radar cross-section of the 159 SL-8 bodies (4.34 $\mathrm{m^2}$) as a proxy for the mean projected area, $C_d = 2.2$ (A) and 1,435 kg (A). A single burn from 975 km to perigee h_p costs $\Delta V = v_c - v_a(h_p)$. The 25-yr rule needs $h_p = 417$ km (371–466 km over the density bracket), 146 m/s and 81 kg of propellant at $I_{sp} = 280$ s; the 5-yr rule needs 314 km, 175 m/s and 97 kg; and a controlled re-entry with a 50 km perigee needs 251 m/s

Table 8: Servicer Mass Allocation and Launch Mass

	chemical carrier	electric carrier
RPO suite, capture and kit interfaces (27.4%)	192	247
structure (21.3%)	150	192
thermal (3.6%)	25	32
power (31.9%)	224	288
TT&C (4.8%)	34	43
ADCS (6.9%)	48	62
propulsion (3.8%)	27	34
dry mass (A)	**700**	**900**
16 kits, dry 135 kg each (A)	2,160	2,160
kit rendezvous propellant	210	210
carrier propellant	212	39
launch mass, draft kits	**3,282**	**3,309**
launch mass, kits sized for 25-yr life, 5-yr life, re-entry	4,810; 5,099; 5,854	–

kg. Fractions: average of SMAD Table A-2 (communication satellites), normalised; masses from the rocket equation with the ΔV of Section 9.

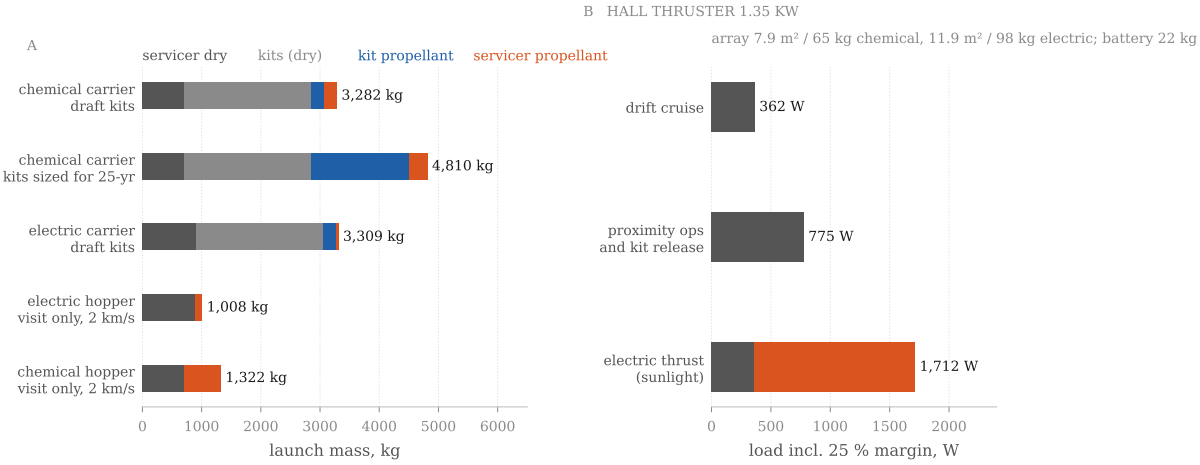


Figure 14: Launch mass by design and electrical load by mode. The kits, not the servicer, set the launch mass; electric thrust more than doubles the array power. Loads are assumptions (Appendix B); sizing after SMAD Sec. 11.4.

and 137 kg. The de-orbit propellant alone is therefore 60–100% of the draft's kit mass, and with it each kit weighs 237–298 kg at departure. For 16 kits the carrier's launch mass rises from 3.3 t to 4.8–5.9 t (Table 8), a factor of 1.47–1.78. The time advantage of the carrier is unchanged, but its mass advantage over a hopper (Table 6) shrinks accordingly, and the kit becomes a small spacecraft of its own with propulsion, avionics, navigation and a capture device.

10.8 Reliability over 7–15 years

For a series system $R = e^{-\lambda t}$, and n identical parallel strings give $R = 1-(1-R_1)^n$ [35, Eqs. (19-1), (19-3), (19-6), pp. 765–766]. No sourced failure rate for such a servicer was found, so we bracket $\lambda = 0.02$–0.2 per year (A). A single string then survives 7 yr with probability 0.87–0.25 and 15 yr with 0.74–0.05; dual strings raise this to 0.98–0.43 and 0.93–0.10. SMAD's data show that complex, programmable and actively controlled subsystems fail several times more often than simple ones [35, Table 19-4, p. 769], which is the case for an autonomous servicer. The architecture matters more than the redundancy. Since a failure ends all later placements, the

expected number of kits placed is $\sum_k e^{-\lambda t_k}$. For the median start, the carrier places 15.9, 15.7, 15.5 and 15.0 of its 16 kits at $\lambda = 0.02$, 0.05, 0.1 and 0.2 per year, and the serial hopper 15.5, 14.7, 13.6 and 11.7. The 36-object visit tour of 6.6 years delivers 33.7 to 19.9 expected visits over the same range. Short campaigns are the most effective reliability measure for this mission.

10.9 Mission timeline

Figure 15 shows the timeline for the median start. After an assumed development of about five years (the value used by [49]) and 90 days of commissioning (A), the chemical carrier descends in about an hour and releases its kits as the drift brings each plane into line, the 16th at about 310 days after launch. An electric carrier releases the same kits about 150 days later, and a hopper that carries them needs about 1,320 days. After the last release the carrier must itself be disposed of: its parking orbit at 573 km has a natural life of about 17 yr in our model, which meets the 25-yr rule but not the 5-yr rule, for which a 36 m/s burn to a 442 km perigee is needed (149 m/s for a controlled re-entry).

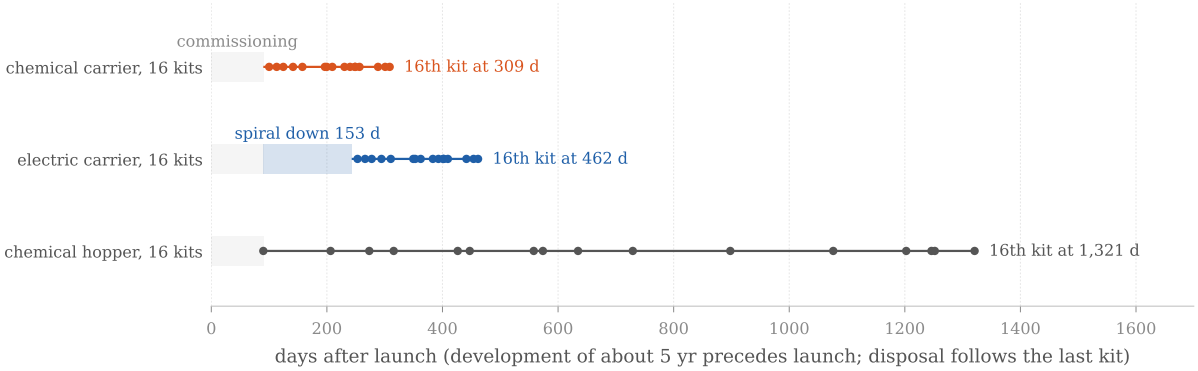


Figure 15: Kit release times after launch for the median start. The parked chemical carrier is done in about 310 days; the spiral of an electric carrier and the serial hopper add 150 and about 1,000 days. Commissioning (90 d) and development are assumptions.

10.10 Cost estimate

The cost was estimated with the parametric relations of SMAD Chapter 20, in fiscal-year 2000 dollars and converted to 2024 dollars with the consumer price index (factor 1.822 from annual averages of 172.2 and 313.7 [55]). The method is as follows. The bus (dry mass minus the 27.4% payload share) uses the large-satellite cost relations, 101 X for development and 43 X for the first unit, with X the dry mass in kg and standard errors of 33% and 36% [35, Tables 20-4, 20-5, pp. 795–796]; the RPO and capture payload is taken as 0.4 times the bus, the small-satellite convention [35, Table 20-6, p. 797], applied here beyond its range. Integration, program level, ground support equipment and launch support follow the same tables. Each kit is costed as a 135 kg small satellite ($781 + 26.1X^{1.261}$ k$, standard error 3,696 k$), split 60/40 into non-recurring and recurring cost [35, Table 20-9, p. 799], with production of N units costing the first unit times $L = N^B$, $B = 1 - \ln(100/S)/\ln 2$, and a 90% slope for 10–50 units [35, Eq. (20-5), p. 809]. Software costs 435 k$ per thousand lines in flight and 220 on the ground (60 and 40 thousand lines, A), the ground segment is three times its software [35, Tables 20-10, 20-11, pp. 800–801], and operations cost 160 k$ per staff-year plus 10% of the ground equipment per year [35, Table 20-12, p. 801]. Launch is priced at a published rideshare list price of 350,000 USD for 50 kg plus 7,000 USD/kg [45] (secondary, 2026 dollars, not inflated). The uncertainty of each element

combines its standard error with a technology term from its readiness level, $\sigma = (\sigma_c^2 + \sigma_t^2)^{1/2}$ [35, Eq. (20-1), p. 805], with TRL 7 for the bus and 4 for the capture payload and kits ($\sigma_t = 10$ and 20%, [35, Table 20-15, p. 804]); the system σ is the average of the fully correlated sum and the uncorrelated root sum square [35, p. 805]. Table 9 gives the result.

Table 9: Parametric Cost Estimate, Chemical Carrier with 16 Kits (2024 M USD)

Element	cost	σ (%)
bus development / first unit	93.4 / 39.8	34 / 37
RPO and capture payload, development / first unit	37.4 / 15.9	39 / 41
integration, assembly and test	43.2	47
program level	62.4	40
ground support equipment	22.1	35
launch and orbital operations support	6.2	43
kit development	14.7	34
kit production (16 units, 90% learning)	102.9	34
flight software / ground segment	47.5 / 48.6	34 / 34
operations (1.3 yr)	9.0	32
launch (3,282 kg, rideshare list price)	23.0	–
total (system σ 133 M; 10–90%: 396–736 M)	**566**	23
serial hopper with 16 kits (3.6 yr of operations)	590	23
electric visit-only hopper, 2 km/s, 7 yr	559	25

Management reserve (about 20% [35, p. 803]) not included. Kits sized for re-entry add 11–18 M of launch cost.

The estimate is about 566 M USD (10–90%: 396–736 M), or 35 M per object and about 24,700 USD per kilogram of rocket body removed. Three points follow from Table 9. First, development dominates: the operations and launch that distinguish a carrier from a hopper differ by only 24 M, so the faster carrier is also no more expensive. Second, the 16 kits cost about a fifth of the total, and they are the element with the least heritage. Third, a visit-only tour costs about as much as a 16-kit removal mission and removes nothing. For comparison, the NASA cost-benefit study gives 3,000–40,000 USD per kilogram for a tug to uncontrolled re-entry [10] (dollar year not stated), ClearSpace-1 costs about EUR 100 M for one 95 kg object [14], and RemoveDEBRIS cost EUR 15.3 M for a demonstration on its own targets [13]. The estimate rests on cost relations derived from satellites of the 1990s, and the capture payload and kits have no direct analogue, so it is a cost class (several hundred million dollars) rather than a price.

11 Regulatory and Policy Context

The rules that govern disposal apply to the servicer and its kits, not retroactively to the targets. The IADC guidelines (Rev. 4, 2025) ask that LEO objects be removed by re-entry or moved to an orbit with a residual life “as short as practicable and no more than” 25 years, with a disposal success probability of at least 90% and a goal of 99% [22]; the UN COPUOS mitigation guidelines (2007) and long-term sustainability guidelines (2019) carry the same intent [51, 52], and ISO 24113:2023 is the top-level standard [23]. Two stricter rules now apply in practice. The US FCC requires disposal no later than five years after the end of the mission for satellites it authorises that are launched after 29 September 2024 [18, 19], and ESA’s requirements of 2023 set an orbital lifetime below five years, a disposal probability above 0.9 and a collision-avoidance threshold of 10^{-4} per conjunction [12]. For this concept, the kits sized above meet either rule for the targets, and the carrier needs a 36 m/s burn to meet the five-year rule. A controlled re-entry would remove the casualty risk on the ground, for which the IADC limit is 10^{-4} per re-entry [22]; we did not compute the casualty area of an SL-8 body.

Two further constraints are legal and operational. The state of registry retains jurisdiction and control over its objects [53], and 158 of the 159 SL-8 bodies carry one owner code in the catalogue, so a removal campaign in this band needs that state's consent or participation. Damage in space is a matter of fault [54], which makes the close approaches of the servicer a liability exposure for its operator. The drift orbits also sit in the lower LEO shells, down to about 275 km, for months at a time, so the servicer must take part in routine conjunction assessment: screening against the public catalogue and conjunction messages, a manoeuvre threshold of the order of 10^{-4} as used by NASA's conjunction assessment practice [44] and ESA [12], and coordination through the civil traffic-coordination service that the US Office of Space Commerce plans to run in parallel with military services [47]. The planned close approaches to each target are themselves conjunctions and should be announced in the same way.

12 Discussion

12.1 Implications for multi-object removal

The physical core is simple. The nodal rate is a steep function of altitude, so a drop of 25–150 km buys rate differences of 0.01–0.06°/day for 25–150 m/s round trip, and the node of a prograde orbit at 83° then closes a few-degree gap in 30–300 d. The numbers add five points to the draft. First, the size of the advantage over impulsive plane changes is a function of the time allowed, and it is large only if years of bus life are available. Second, the benefit of choosing the sign per leg is real (about two objects, and a factor of two in expected wait), so the unidirectional rule is not required. Third, the sequencing heuristic was not the weak point; the scalarisation weight and the best-of-N reporting were. Fourth, the visit count is not a removal count. With 135 kg kits, eight kits imply about 2.1 t at launch for a hopper, so removals per launch are mass-limited well below 29 or 67 objects, which is why the carrier-and-kits architecture is the one idea that matters. Fifth, in this model the collision-risk benefit of a 28-object removal from SL-8 is a few hundredths of a catastrophic collision over 25 years. The case for removal therefore rests on the heavy tail and on cascade dynamics, not on the first-generation average, and the model cannot settle it.

The servicer design of Section 10 adds three points. First, the time that drift phasing costs is also a reliability cost, and it is the strongest argument for the parked carrier: placing 16 kits in about 310 days keeps the expected loss below one kit even at a failure rate of 0.2 per year, while a serial hopper loses about four. Second, the mass and cost of a removal mission are set by the kits. Sized to lower a 1.4 t body, each kit needs 81–137 kg of de-orbit propellant on top of the draft's 135 kg, and the 16 kits are about a fifth of the cost. Third, at about 35 M USD per object the concept is below the cost per object of ClearSpace-1 (about EUR 100 M for one object), but it is not yet credible as a plan, because the capture of a tumbling, unprepared rocket body by a small autonomous kit has not been demonstrated by any mission.

12.2 Novelty assessment

Already known. Using J_2 drift for multi-debris tours, with drift orbits below, between or above the debris, is in Cerf (2013), with a lower-orbit worked example [6, 7]. The GTOC9 winners exploited it in branch-and-bound and beam searches [25, 48]. Raised waiting orbits with deployable kits are in [49], and a carrier with 2–8 kits waiting in a drift orbit 100 km below is in [9]. That altitude-lowering is more efficient than raising for nodal phasing is in [42], and the principle is patented prior art from 1993 [32]. Drift-orbit optimisation with drag in servicing is in [2], and with low thrust in [30]. Time-dependent TSP, evolutionary multi-chaser and dynamic-

programming sequencing are in [20, 26, 41, 56]; risk-based ADR planning exists [21, 46]; and a recent preprint treats the drift orbit as a design variable and finds that inclination tilt beats altitude for near-sun-synchronous targets [24].

Not found in print (elementary, apparently unpublished). The closed-form ratio $R(x)$ with its remainder bound, the $1 + \frac{3}{2}x$ Hohmann ratio and $1 + 3x$ net gain, the decision threshold $180°(1 + \frac{9}{4}x)$ and the eccentric-target reversal. These follow from the convexity of $a^{-7/2}$ and small-parameter series, so their weight is small.

New evidence, not a new method. The audit and frozen snapshot; the weight correction and exact-optimum validation; the Cowell/SGP4 closure test; the quantified baselines; the risk benefit with its uncertainty; the novel-idea screen (ten ideas); the exact reproduction of the 2025 counts; and a subsystem-level servicer design with a cost estimate.

Candidate novelty. The persistent-drift-orbit carrier whose kits depart without a carrier return, quantified here but untested against a full kit design. We do *not* claim new astrodynamics, any removal capability, or any prediction of environment evolution.

12.3 Sources of error and model limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

Leg model: The leg model is impulsive and circular-to-circular and has no in-plane phasing. Proximity operations and capture are one fixed allowance (15 m/s), which is unverified and decisive: an allowance of 50 m/s costs −8 objects.

Free insertion: Launch inserts the chaser at the start object for free, whereas a real launch inserts at its own plane and node.

Element-set errors and verification sample: Element-set errors beyond the assumed values are untreated, and the verification covers eight legs of one tour.

Atmosphere: The atmosphere is a tabulated exponential model (unverified, uncertain by a factor of 2–3). Solar activity is bracketed, not simulated.

Mass and kit assumptions: The mass and kit assumptions (700 kg, 135 kg, I_{sp}) are stated, not sourced.

Risk model: The risk model is a kinetic-gas estimate with 66% imputed cross-sections, no cascade feedback and no operational-asset weighting.

Pools and epoch: The pools are four bands of a 2026-08-18 catalogue. Results for other bands (especially near 600 km, where drag matters) are not claimed. No re-fetch of the catalogue was made, so the numbers describe that snapshot only.

Relaxation bound: The relaxation bound on the full pool is loose, so the full-pool optimality gap is not certified.

Literature coverage: The literature coverage is a bounded search, and items marked unverified in `research/sources.md` were not used.

Servicer budgets: The loads, the dry masses and the mass fractions are assumptions or communication-satellite averages, not a design. They place the servicer in the right class (hundreds of watts, about a tonne dry) and show that the kits dominate, but a heavier capture payload would raise the dry mass and, through the cost relations, the development cost roughly in proportion.

Kit sizing and lifetime model: The residual-life perigees use an exponential atmosphere, a radar cross-section as a proxy for area and a single burn; the density bracket moves the 25-yr

perigee by ±45 km and the propellant by ±7 kg, so the conclusion that the draft's kit is too light holds. Kit avionics, capture hardware and tanks are still not counted.

Reliability: No sourced failure rate exists for this class of servicer, so the reliability results are a bracket over $\lambda = 0.02$–0.2 per year. The ranking of carrier over hopper holds over the whole bracket; the absolute values do not.

Cost: The cost relations were derived from 1990s satellites and are applied to a payload and to kits without analogues; the small-satellite payload factor is used beyond its range, the software sizes and staffing are assumptions, and the launch price is a secondary list price. The 10–90% interval reflects the published standard errors and readiness levels, not these model choices, so the true uncertainty is larger. The result is a cost class, and it should be compared with the demonstrations only at that level.

12.4 What a reviewer would challenge

The strongest objections to this analysis, and our answers, are as follows.

1. *"Your gain over the draft is a weight change."* Yes: the draft's configuration was untuned. The comparison against the *published-style* fixed orbits (Table 3) is the relevant one for a literature claim.
2. *"The impulsive baseline is a straw man."* It is the best single-burn hop with an exact Hohmann and inclination charge, but it excludes multi-burn bielliptic options (tested and worse for small angles) and a plane change at higher altitude. Its time (16–36 d) is its real advantage, and a time-weighted objective could reverse the ranking.
3. *"15 m/s proximity operations and no phasing."* Correct, and decisive (Section 8). The count is conditional on that allowance.
4. *"Free insertion at the start object."* Yes; the starting-plane acquisition is not charged.
5. *"Counts are visits, not removals."* Yes; see Section 9.
6. *"Single pool, single epoch."* The pool is nearly exhausted at 5 km/s, and the SSO, Iridium-class and SL-16 bands appear in Table 4 only as medians.
7. *"Optimality gap not certified on the full pool."* Correct (Section 6).
8. *"Atmosphere, mass, kit and cost values are assumptions."* Yes. They are tagged in `research/params.json` (VERIFIED/DERIVED/UNVERIFIED); the headline numbers are insensitive to the atmosphere for this pool and sensitive to the proximity allowance.
9. *"The risk model is not validated."* It has no calibration to observed conjunction or fragmentation rates beyond the sanity check of 55 years between events if every derelict pair were involved, so it should be used for ranking only.
10. *"The cost estimate is a parametric guess."* Yes. It is a SMAD-style estimate with stated inputs and uncertainty, adequate to place the concept in a cost class and to show that development, not operations, dominates (Table 9).
11. *"Was the literature search complete?"* No. Several papers named in the brief (Han 2024, Yang 2024 as stated, Ruggiero, Lüthi) could not be verified and were not used.

13 Conclusion

This work re-evaluated nodal-precession phasing for multi-object debris removal on one frozen catalogue snapshot, reproduced the draft study's 2025 pipeline from its original inputs, checked

the planner against an exact optimum and the leg model against high-fidelity propagation, and turned the tour into a servicer design with a cost estimate and its regulatory context.

At the level of the phasing geometry, lowering gains over raising by $R(x) = 1 + \frac{9}{2}x + \dots$ in rate and by $1 + 3x$ per unit ΔV. The gain is small (4% at 100 km) and one-signed, so the bidirectional rule is better. At the level of the planner, a tuned beam search is within 2% of optimal on small instances. At the level of the data, the draft's 63,553, 1,288 and 593 are reproduced exactly; they come from two different 2025 catalogues and include objects that had already re-entered, and the comparable 2026 count of objects in orbit is 1,151.

At the mission level, a single spacecraft can visit a median of 29 SL-8 rocket bodies in 7 yr with 2 km/s, and 67 in 15 yr with 5 km/s, against 9 and 19 by impulsive plane change. Visits are not removals. The architecture that turns drift phasing into removal is a chemical carrier parked below the cluster that releases one kit per object; it places 16 kits in about 310 days, which also makes it the most reliable option, but kits that can actually lower a 1.4 t body make it a 4.8–5.9 t launch and a mission in the several-hundred-million-dollar class, about 35 M USD per object. Every flown or planned demonstration so far handles one object, and none has captured a tumbling, unprepared rocket body.

In general, the results demonstrate that drift phasing trades mission time for propellant rather than providing a time-free saving, and that its practical value lies in the number of objects per launch, which is limited by the kits rather than by the orbital mechanics. The main recommendations are as follows:

1. Plan removal around a parked carrier with kits and short campaigns, not a serial tour; the carrier is faster, more reliable and no more expensive.
2. Use chemical propulsion for the carrier and the kits; keep electric propulsion for visit-only inspection tours, with at least two thrusters.
3. Size the kit first: its de-orbit propellant (81–137 kg for an SL-8 body), its capture device and its autonomy set the launch mass and a fifth of the cost.
4. Fly a single-kit demonstration on one SL-8 body, combining the non-cooperative rendezvous already shown in orbit with a kit attachment and a perigee-lowering burn, before any multi-object campaign.
5. Report counts on a dated, hash-pinned catalogue with an explicit decay filter, and report tours as distributions over starts rather than as the best case.
6. Treat consent of the state of registry, conjunction coordination and the five-year disposal rule as design inputs from the start.

Future work should model the in-plane phasing and proximity-operations costs explicitly, design the kit and its capture mechanism, compute the re-entry casualty risk of an SL-8 body to decide between controlled and uncontrolled disposal, and extend the analysis to other bands and to a refreshed catalogue.

Acknowledgments

The author acknowledges the public satellite catalogue and element-set services that made the frozen snapshot possible, and the authors of the open-source propagation and scientific Python libraries used throughout this analysis.

A Reproducibility Statement

Snapshot. v3-2026-08-18: Celestrak SATCAT csv (70,292 rows) and GP three-line sets (SL-8 159, SSO 135, Iridium-class 28, SL-16 20, mixed 393), public, no credentials, fetched 18 August 2026; the SHA-256 of every file is in `data/SNAPSHOT.json`.

Code. `run_all.py` runs the tests, audit, analysis, figures and this PDF; version hash `68c3af00e0f40e7a` over 101 files (`CODE_MANIFEST.json`); Python 3.14, numpy 2.4, scipy 1.17, matplotlib 3.10, sgp4 2.25, sympy 1.14; no GPU, about 80 min on one core. This version was prepared on 4 November 2026.

Seeds. Start subsets 7, 9, 42; Monte Carlo 1, 2026, 11, 100+replicate; NSGA-II 0–2; bootstrap 1–2; every stochastic step is seeded inside its script.

Provenance. The original repository was read-only; its code is copied to `legacy/` and used only to reproduce its numbers. No data were invented, and unverified inputs are listed in `research/params.json`.

Version 4 additions. `analysis/a9_orig2025.py` reproduces the 2025 counts from `data/orig2025/` (Section 3.1); `analysis/a8_servicer.py` computes the servicer budgets, trade grid, reliability, kit sizing, timeline and cost (Section 10); `report/make_figures_v4.py` draws Figs. 12–15 for the paper and the site versions in `figs_site/` with their `manifest.json`. The mission data and their status (verified or secondary) are in `research/missions_v4.json`.

B Servicer Parameters

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Table 10: Inputs of the Servicer Design (M: Measured or Published, D: Derived, A: Assumed)

Parameter	Value	Tag	Source
dry mass, chemical / electric servicer	700 / 900 kg	A	draft design, `research/params.json`
specific impulse, chemical / electric / kit	320 / 1,800 / 280 s	A	within SMAD Table 17-4, p. 692
Hall thruster power, thrust, demonstrated life	1.35 kW, 83 mN, >5,700 h	M	SMAD Table 17-11, p. 707
kit dry mass	135 kg	A	draft concept note
SL-8 body mass	1,435 kg	A	`research/params.json`
SL-8 area proxy (median radar cross-section, 159 bodies)	4.34 m^2	M	2026 catalogue snapshot
drag coefficient	2.2	A	–
loads (cruise / proximity operations), before margin	290 / 620 W	A	guided by SMAD Table 10-9, p. 316
load margin	25%	M	SMAD Table 10-9, p. 316
cell efficiency, degradation	22%, 0.5%/yr	M	SMAD Table 11-34, p. 412
inherent degradation, Sun angle	0.77, 23.5°	M	SMAD Tables 11-34, 11-36, pp. 412–414
path efficiencies X_e, X_d	0.60, 0.80	M	SMAD p. 413
battery DOD, specific energy, efficiency	25%, 90 W h/kg, 0.90	A/M	SMAD pp. 420–422 (15–25%, 70–110 W h/kg)
array specific power	25 W/kg	M	SMAD Eq. (10-13), p. 333
subsystem mass fractions	Table 8	M	SMAD Table A-2, p. 895
failure rate bracket	0.02–0.2 /yr	A	no sourced value
commissioning; development	90 d; about 5 yr	A	[49] for development
flight / ground software	60 / 40 thousand lines	A	–
operations staff	12 staff-years per year	A	–
cost relations, learning slope, TRL sigma	as in text	M	SMAD Tables 20-4 to 20-15, pp. 795–809
TRL: bus / capture payload and kits / software	7 / 4 / 5	A	our reading of SMAD Table 20-15
CPI-U 2000, 2024 annual average	172.2, 313.7	M	[55]
rideshare list price	350,000 USD + 7,000 USD/kg	M	secondary [45]
ground station mask	10°	A	–

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